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Prevalence and Association of Atherosclerosis to Ischemic Stroke in Patients With Atrial Fibrillation on Anticoagulation

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BACKGROUND: Approximately, 20% of ischemic strokes in patients with atrial fibrillation occur despite anticoagulation. These breakthrough strokes are associated with a high risk of recurrence, but underlying mechanisms remain incompletely understood. In particular, the association between carotid atherosclerosis and breakthrough strokes has not been sufficiently assessed.

METHODS: We analyzed data from the prospective, multicenter, RASUNOA-Prime cohort study (Registry of Acute Stroke Under Novel Oral Anticoagulants–Prime), conducted at 46 German stroke centers between 2015 and 2020. Eligible patients had atrial fibrillation and an ischemic stroke within 24 hours before admission. Of 2737 patients, computed tomography angiography was available for 1464 (53.5%). Patients were grouped according to prestroke anticoagulation with direct oral anticoagulants, vitamin K antagonists, or no oral anticoagulation. Carotid atherosclerosis, including stenosis or occlusion and nonstenosing vulnerable plaques, was assessed by core laboratory computed tomography angiography readings. Carotid arteries served as observational units, with presence of atherosclerosis as binary outcome and laterality of carotid-territory ischemia as explanatory variable in generalized linear mixed models. Laterality of atherosclerosis was assessed as ipsilateral if present on the side of carotid-territory ischemia.

RESULTS: Any carotid atherosclerosis was identified in 81% of 1464 patients with available computed tomography angiography. Extracranial carotid stenosis $\geq 50\%$ or occlusion was present in 17%. Among 792 patients with unilateral carotid-territory ischemia and no stenosis, ipsilateral vulnerable carotid plaques were detected in 34% (no oral anticoagulation 28%, direct oral anticoagulants 38%, vitamin K antagonists 38%), including patients with bilateral plaques, whereas 5% had contralateral vulnerable plaques only. In direct oral anticoagulant–treated patients, odds of ipsilateral vulnerable plaque were significantly higher than in nonanticoagulated patients (odds ratio, 4.4 [95% CI, 1.6–11.8]; $P=0.004$).

CONCLUSIONS: Stenosing and nonstenosing carotid atherosclerosis represents a frequent comorbidity in patients with breakthrough strokes, which may contribute to their high risk of recurrence. Longitudinal studies including advanced vascular imaging are needed to clarify the impact of atherosclerosis on stroke recurrence after breakthrough strokes in patients with atrial fibrillation.

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GRAPHIC ABSTRACT: A [graphic abstract](#) is available for this article.

Key Words: angiography ■ atherosclerosis ■ atrial fibrillation ■ carotid artery diseases ■ ischemic stroke ■ tomography, x-ray computed

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Nonstandard Abbreviations and Acronyms

AF	atrial fibrillation
CTA	computed tomography angiography
DOAC	direct oral anticoagulant
LDL	low-density lipoprotein
Non-OAC	no oral anticoagulation
VKA	vitamin K antagonist

Ischemic stroke is a major cause of disability and mortality worldwide. Atrial fibrillation (AF) represents a major risk factor for stroke, and oral anticoagulation substantially reduces the risk of thromboembolism in these patients. Nevertheless, ischemic strokes that occur despite oral anticoagulation—also referred to as breakthrough strokes—account for about 20% of all AF-related strokes.^{1,2} Patients experiencing breakthrough strokes face a particularly high risk of stroke recurrence of 5% to 9% per year.^{2,3} Currently, there is no effective antithrombotic strategy for secondary prevention in these patients as switching anticoagulants or adding antiplatelet therapy did not effectively prevent recurrent events.⁴ Hence, the development of targeted prevention strategies for breakthrough strokes has been identified as an important unmet need.⁵ At present, however, the mechanisms underlying breakthrough stroke are not well understood. In a landmark retrospective multicenter study, 32% of breakthrough strokes were attributed to medication errors and 44% to cardioembolism, despite adequate anticoagulation.⁶ The remaining 24% were believed to result from competing mechanisms, with large artery atherosclerosis identified as the most common cause.⁶ In subsequent studies, competing mechanisms accounted for 20% to 35% of breakthrough strokes.^{7,8}

At present, the prevalence of atherosclerosis and the potential contribution of atherosclerotic mechanisms to the pathogenesis of breakthrough strokes may be underestimated. Previous studies focused on the presence of carotid stenosis and did not consider vulnerable carotid plaques which are increasingly recognized as sources of embolic stroke.^{9,10} The Carotid Plaque Reporting and Data System classification system provides a standardized framework to evaluate carotid plaque morphology and composition across multiple imaging modalities with incremental risk of stroke over stenosis degree only.^{10–15} In patients without AF, computed tomography angiography (CTA) studies have shown that features such as soft-plaque thickness >3 mm, the rim sign, and plaque ulceration are associated with ipsilateral ischemic stroke.^{16–23} To date, a detailed prospective analysis of stenosing and nonstenosing carotid atherosclerosis in patients with breakthrough strokes in comparison to AF patients without anticoagulation has not been performed.²⁴

Herein, we present an analysis of CTA scans in a large prospective observational study of ischemic strokes occurring in patients with AF who were either anticoagulated or not at the time of the event. We aimed to determine the prevalence of stenosing and nonstenosing atherosclerosis in patients with breakthrough stroke compared with patients without anticoagulation. Moreover, we examined the association of vulnerable carotid plaques and carotid stenosis with ipsilateral downstream brain ischemia.

METHODS

Data Availability Statement

The data that support the findings of this study, as well as the analytic methods and study materials, are available from the corresponding author on reasonable request. The corresponding author is responsible for maintaining data availability and will provide access in accordance with institutional regulations and applicable data protection laws.

Ethics Statement

The ethics committee of the Medical Faculty of the University Heidelberg as well as all local ethics committees approved the study protocol (S-088/2015). Patients were eligible for the study if they were 18 years or older and either the patient or a legal representative had provided consent. Informed consent could be waived if patients were not able to consent and died before a legal representative could be installed.

Study Design, Setting and Standard Protocol Approvals

The RASUNOA-Prime study (Registry of Acute Stroke Under Novel Oral Anticoagulants–Prime) was a prospective, multicenter, observational study of patients with AF and acute ischemic stroke, that was conducted between 2015 and 2020 in 46 certified German stroke units.^{25,26} Patients aged 18 years or older with a diagnosis of acute ischemic stroke, defined by either neurological symptoms lasting ≥ 24 hours or imaging evidence of acute infarction, were eligible for inclusion. Participants were stratified by anticoagulation status prestroke: (1) no anticoagulation (2) anticoagulation with a direct oral anticoagulant (DOAC; 3) anticoagulation with a vitamin K antagonist (VKA). Patients were eligible for the study if they were 18 years or older and either the patient or a legal representative had provided informed consent. Informed consent could be waived if patients were not able to consent and died before a legal representative could be installed. In the present analysis, only patients who had undergone clinically indicated CTA were included. Brain and vascular images were centrally collected as part of the study protocol.

The ethics committee of the Medical Faculty of the University Heidelberg as well as all local ethics committees approved the study protocol (S-088/2015).

Vascular Imaging Analysis

Indication for cervicocranial CTA was based on local clinical decision making by treating physicians in the hyperacute

phase. Images were later transferred to the neuroradiologic core laboratory for analysis using OsiriX Pro software (version 12 or higher). Two independent, trained readers performed the image review according to a prespecified manual. Readers were blinded to all clinical data and anticoagulation prestroke. In case of disagreement, a third reader was consulted, and final consensus was reached under the supervision of a senior neuroradiologist. Analyses were performed on multiplanar reconstructions in axial, coronal, and sagittal planes. The presence and degree of carotid artery stenosis was determined using the NASCET (North American Symptomatic Carotid Endarterectomy Trial) methodology. Degree of stenosis was categorized between 50% and 70%, 71% and 99%, and occlusion.

Carotid Plaque Features

We classified carotid plaques as vulnerable if they exhibited any of the following features (Figure S1): presence of a rim sign, a soft-plaque thickness >3 mm, an ulcerated plaque surface, or evidence of intraplaque hemorrhage.^{20,22,23,27} A rim sign was diagnosed in the presence of an adventitial, calcified rim with high-attenuation (>130 Hounsfield units) with a diameter smaller than 2 mm, surrounding a low-attenuation plaque core.^{23,27} Soft plaques had a density of <60 Hounsfield units.²⁰ Plaque surface morphology was classified as smooth, irregular, or ulcerated in the extracranial internal carotid artery according to established criteria.²⁸ Intraplaque hemorrhage was defined as a noncalcified plaque component with attenuation values lower than adjacent muscle, typically <25 to 30 Hounsfield units, corresponding to hypoattenuating areas within the plaque.^{29,30} Spotty calcification was defined as small, calcified deposits <3 mm in length along the vessel axis without substantially affecting the vessel lumen and with a circumferential arc of the vessel wall <90°.³¹

Localization of Cerebral Ischemia

The location of cerebral ischemia in the carotid territory was determined if any of the following 3 criteria were met: (1) Intracranial vascular occlusion downstream from the carotids on CTA; (2) acute infarct in the carotid territory on cranial CT; (3) or acute infarct in the carotid territory on brain magnetic resonance imaging.

Statistical Methods

All statistical analyses were performed using R version 4.5.0. Baseline characteristics were summarized using descriptive statistics. Continuous variables are reported as mean±SD and median with interquartile range, whereas categorical variables are presented as counts and percentages. If present, missing or not assessable values are shown in square brackets within the Tables 1 and 2.

To investigate the association between vulnerable plaques and ischemia, generalized linear mixed models with a binomial logistic link function were fitted using the lme4 package.³² Because all included patients experienced an ischemic event, we used a side based analytical approach and considered each carotid artery as an observational unit. The presence of carotid plaque features was modeled as binary outcome, and the laterality of cerebral carotid-territory ischemia (ipsilateral versus contralateral) was used as the main explanatory variable. From a statistical perspective, this approach is equivalent to assessing whether, within the same patient, arteries on the ischemic side are more likely to harbor vulnerable plaque features or significant stenosis than arteries on the nonischemic side. The resulting odds ratios (OR) quantify the strength of the local association between carotid pathology and downstream ischemia.

Table 1. Main Characteristics of the CTA Cohort Stratified by Anticoagulation Status

Variable	All CTA (N=1464)	Non-OAC (n=560)	DOAC (n=522)	VKA (n=382)
Age, y	79 (74–84)	78 (70–84)	80 (75–84)	81 (76–85)
Female	724 (49%)	298 (53%)	246 (47%)	180 (47%)
BMI	27 (24–30) [439]	27 (24–30) [185]	26 (24–29) [145]	26 (24–29) [109]
Baseline NIHSS score	7 (3–15) [61]	8 (4–16) [25]	6 (3–13) [21]	7 (3–15) [15]
mRS score at admission	4 (2–5) [14]	4 (3–5) [4]	3 (2–4) [7]	4 (3–5) [3]
CHA ₂ DS ₂ -VASc	4 (3–5) [10]	4 (3–5) [2]	4 (3–6) [4]	4 (4–6) [4]
Prior stroke/TIA	229 (16%) [3]	56 (10%)	111 (21%) [1]	62 (16%) [2]
Prior ICH	15 (1%) [7]	7 (1%) [2]	4 (1%) [3]	4 (1%) [2]
Arterial hypertension	1313 (90%) [1]	481 (86%)	473 (91%) [1]	359 (94%)
Diabetes	428 (29%) [7]	135 (24%) [2]	148 (29%) [3]	145 (38%) [2]
Hyperlipidemia	596 (42%) [35]	200 (36%) [10]	227 (45%) [17]	169 (45%) [8]
Ischemic heart disease or prior myocardial infarction	442 (30%) [5]	136 (24%) [2]	174 (33%) [1]	132 (35%) [2]
Smoking	170 (12%) [28]	66 (12%) [6]	62 (12%) [12]	42 (11%) [10]
Alcohol abuse	73 (5%) [11]	35 (6%) [2]	21 (4%) [5]	17 (4%) [4]
Antiplatelet therapy	297 (21%) [41]	220 (40%) [15]	53 (10%) [12]	24 (7%) [14]

Categorical variables: n (%); continuous variables: median (IQR). Missing values, if present, are shown in square brackets. All statistics refer to valid n. NIHSS: from 0 (no symptoms) to 42. mRS score: from 0 (no symptoms) to 6 (death). CHA₂DS₂-VASc score: range 0–8, from low to high risk of ischemic stroke in atrial fibrillation. BMI indicates body mass index; CTA, computed tomography angiography; DOAC, direct oral anticoagulants; ICH, intracerebral hemorrhage; IQR, interquartile range; mRS, modified Rankin Scale; n/N, number of patients; NIHSS, National Institutes of Health Stroke Scale; non-OAC, no oral anticoagulants; TIA, transient ischemic attack; and VKA, vitamin K antagonists.

Table 2. Prevalence of Extracranial Carotid Atherosclerosis Phenotypes by Anticoagulation Status

Variable	All CTA (N=1464)	Non-OAC (n=560)	DOAC (n=522)	VKA (n=382)
Any atherosclerosis	1 192 (81%) [13]	449 (80%)	421 (82%) [8]	322 (84%) [5]
Nonstenosing atherosclerosis	938 (64%) [13]	358 (64%)	320 (61%) [8]	260 (68%) [5]
Stenosing/occlusive atherosclerosis	254 (17%) [13]	91 (16%)	101 (19%) [8]	62 (16%) [5]
Unilateral stenosis*	140 (10%) [13]	46 (8%)	62 (12%) [8]	32 (8%) [5]
Bilateral stenosis	28 (2%) [13]	9 (2%)	10 (2%) [8]	9 (2%) [5]
Unilateral occlusion*	96 (7%) [13]	41 (7%)	33 (6%) [8]	22 (6%) [5]
Bilateral occlusion	0 (0%) [13]	0 (0%)	0 (0%) [8]	0 (0%) [5]

All percentages (%) are rounded and relative to all CTA patients (N=1 464). Not assessable values, if present, are shown in square brackets. All statistics refer to valid n. CTA indicates computed tomography angiography; DOAC, direct oral anticoagulants; non-OAC, no oral anticoagulants; and VKA, vitamin K antagonists.

*Subcategories are not mutually exclusive. Ten patients with stenosis on one side and occlusion on the other are counted in both rows.

In this model within-patient interdependence was represented by random intercepts per patient. Besides laterality of the ischemia, age (*Z* scaled), sex, anticoagulation status (no oral anticoagulation [non-OAC], DOAC, VKA), and the interaction between ipsilateral stroke and anticoagulation status were included as fixed effects. Based on these models, group-specific ORs (DOAC, VKA, and non-OAC) for the effect of ipsilateral versus contralateral ischemia on vessel status were derived from estimated marginal means contrasts using the emmeans package. Wald-based 95% CIs and corresponding *P* values were calculated for each contrast. Analyses were conducted both overall and stratified by sex. The same modeling strategy, covariates, and random effects were applied to side specific stenosis or occlusion (yes/no) as binary outcome variable. Missing values in key variables resulted in exclusion of the respective patients from the analysis, as detailed in the patient flowchart (Figure 1). We cannot rule out physiological reasons for missingness and therefore that data are missing not at random.

RESULTS

Patient Flow and Baseline Characteristics

Of 2 737 patients with acute ischemic stroke with AF enrolled in RASUNOA-Prime, 1 464 (53%) had undergone CTA (study flow in Figure 1). Overall, the baseline characteristics of the CTA cohort (Table 1) did not differ substantially from the entire RASUNOA-Prime cohort (Table S1; Table S2). The median age was 79 years (interquartile range, 74–84) and 49% (n=724) were women. The median National Institutes of Health Stroke Scale on admission was 7 (interquartile range, 3–15). The cohort had a substantial prevalence of vascular risk factors (ie, arterial hypertension 90%, diabetes 29%, hyperlipidemia 42%) and 30% had a history of ischemic heart disease or a prior myocardial infarction. A prior stroke or transient ischemic attack was reported in 16%. Before the index stroke, 21% received antiplatelets, and 62% were taking oral anticoagulants (DOAC 36%, VKA 26%). Patients were stratified by anticoagulation status prestroke (Table 1). Patients taking a DOAC or a VKA tended to be older, more often male, and

more likely to have a history of prior cerebrovascular events or vascular comorbidities than non-OAC patients and had a less severe neurological deficit.

Prevalence of Extracranial Carotid Atherosclerosis, Stenosis and Occlusion

In the CTA cohort, 1 192 patients (81%) had any carotid atherosclerosis, and the prevalence was slightly higher in orally anticoagulated patients (Table 2). Carotid stenosis or occlusion was observed in 17% (n=254). Ten patients had a carotid stenosis on one side and a carotid occlusion on the other side. Nonstenosing atherosclerotic features were found in 64% (n=938).

Prevalence of Vulnerable Carotid Plaque Features

Among patients with nonstenosing atherosclerosis, unilateral vulnerable plaques were present in 16% (n=154) and bilateral vulnerable plaques in 34% (n=314; Table S3). Orally anticoagulated patients had a higher prevalence of bilateral vulnerable plaques (DOAC=38%, VKA=35%) compared with not anticoagulated patients (29%). The rim sign was the predominant vulnerable plaque feature, occurring bilaterally in 33% of all patients with nonstenosing atherosclerosis.

Laterality of Extracranial Carotid Stenosis or Occlusion in Patients With Unilateral Ischemia in the Carotid Territory

Among the 933 patients with unilateral carotid-territory ischemia, ipsilateral extracranial carotid stenosis (NASCET ≥50%) or occlusion was identified in 11% (n=105; Table S4). In contrast, contralateral stenosis or occlusion was present in 6% (n=51) of all patients with unilateral ischemia (Table S4). The laterality of stenosis or occlusion appeared similar across anticoagulation groups (Table S5). Generalized linear mixed model

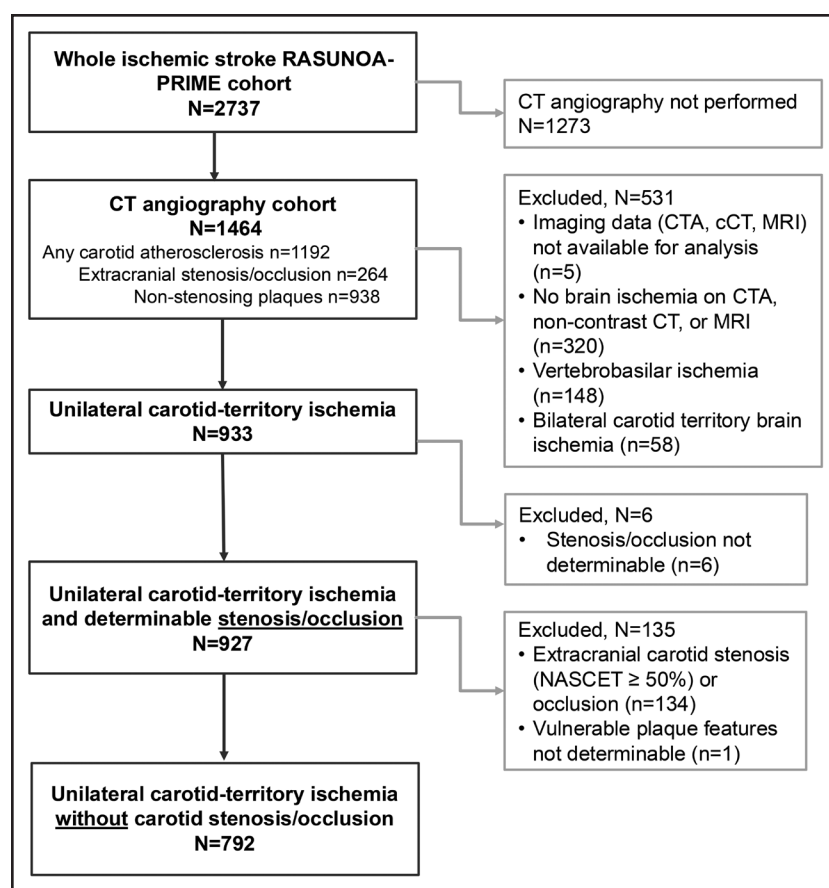


Figure 1. Patient flow of RASUNOA-Prime (Registry of Acute Stroke Under Novel Oral Anticoagulants-Prime) computed tomography angiography (CTA) cohort.

CT indicates computed tomography; cCT, cranial computed tomography; MRI, magnetic resonance imaging; and NASCET, North American Symptomatic Carotid Endarterectomy Trial.

analysis (Figure 2; Table S6) revealed that, among all patients with unilateral ischemia, the odds of having a stenosis or occlusion were substantially higher on the side ipsilateral to the stroke compared with the contralateral side (DOAC: OR, 22.77 [95% CI, 5.90–87.96]; $P<0.001$; VKA: OR, 24.82 [95% CI, 4.20–146.88]; $P<0.001$; non-OAC: OR, 13.47 [95% CI, 4.15–43.71]; $P<0.001$).

Laterality of Extracranial Carotid Plaque and Unilateral Ischemia in Patients With Nonstenosing Atherosclerosis

After excluding 134 patients with extracranial carotid stenosis or occlusion, 6 patients without determinable plaque features and 1 patient without assessable plaque features from the initial cohort of 933 patients with unilateral carotid-territory ischemia, 792 patients remained for the analysis of nonstenosing atherosclerosis laterality. Among these, 208 had vulnerable carotid plaques on both sides and 61 had a vulnerable plaque on the ipsilateral side only, yielding a total of 269 patients (34%) with an ipsilateral vulnerable plaque (Table S7). In generalized linear mixed model analysis (Figure 3; Table S8) of patients with breakthrough strokes taking a DOAC, the odds of finding a vulnerable plaque on the ipsilateral compared with the contralateral side

were increased (OR, 4.37 [95% CI, 1.62–11.80]; $P=0.004$). In contrast, no significant association was observed in patients taking no anticoagulant (OR, 1.29 [95% CI, 0.58–2.88]; $P=0.540$) or a VKA (OR, 1.24 [95% CI, 0.50–3.08]; $P=0.643$). Stratified generalized linear mixed model analyses indicated a strong association for women taking a DOAC (OR, 11.76 [95% CI, 2.03–67.98]; $P=0.006$) while the association did not reach statistical significance in men (OR, 1.55 [95% CI, 0.61–3.91]; $P=0.353$).

DISCUSSION

The major new findings of the present study are that patients with breakthrough strokes have a high prevalence of stenosing and particularly nonstenosing carotid atherosclerosis, and that both stenosis and nonstenosing vulnerable carotid plaques are associated with ipsilateral ischemia in the carotid territory.

Carotid artery atherosclerosis and AF frequently coexist. A meta-analysis including 20 studies with 49 070 patients with AF found carotid plaques in 48%.³³ Conversely, the same study included 28 studies with over 2 million patients with carotid disease and found AF in 9%.³³ One prospective, multicenter cohort study using carotid ultrasound in patients with breakthrough stroke reported evidence of any carotid atherosclerosis in 65%.³⁴

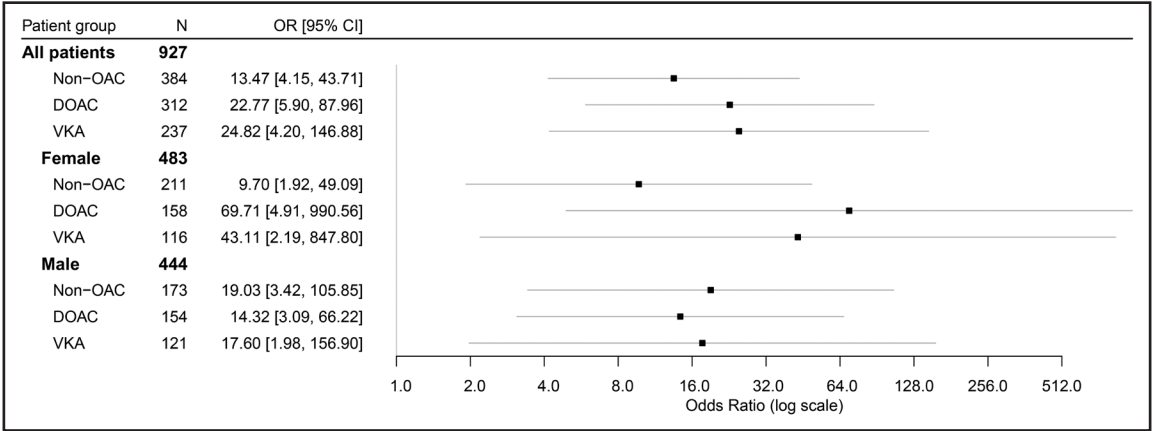


Figure 2. Odds ratios (OR) and 95% CIs for the association between ipsilateral stroke and carotid stenosis or occlusion, estimated using a generalized linear mixed model and group-specific contrasts with emmeans.

To obtain the odds ratios for the direct oral anticoagulants (DOAC) and vitamin K antagonist (VKA) groups, the main effect of stroke ipsilateral (in the reference group no oral anticoagulation [non-OAC]) and the respective interaction effect (stroke ipsilateral×DOAC or ×VKA) were multiplied. Analyses were performed in the overall cohort and stratified by sex.

Our study found any carotid atherosclerosis in 81% of patients with AF, regardless of anticoagulation status.

Hemodynamically relevant carotid stenosis is an established risk factor for ischemic stroke, and symptomatic patients are candidates for endarterectomy or endovascular stenting.³⁵ In contrast, asymptomatic patients might only benefit from intervention in case of high-grade stenosis, as those with moderate stenosis have a low 5-year stroke risk.^{36,37} In a meta-analysis of patients with AF, the prevalence of carotid stenosis ≥50% was 12.4%.³³ Among patients with stroke with AF, concomitant carotid stenosis ≥50% was present in 18% to 24% and independently doubled the risk of both recurrent

stroke and short-term mortality.^{38,39} In our study stenotic or occlusive carotid disease was strongly associated with brain ischemia in the ipsilateral carotid-territory regardless of anticoagulation therapy. Taken together, concomitant carotid stenosis contributes substantially to stroke risk in patients with AF overall which may be particularly relevant in the context of breakthrough strokes.⁴⁰

A meta-analysis of 18304 patients with ischemic stroke or transient ischemic attack reported that 51% had nonstenosing atherosclerosis.⁴¹ In our study, 64% of patients had evidence of nonstenosing atherosclerosis. In recent years, increasing attention has been paid to vulnerable atherosclerotic plaques in nonstenotic carotid arteries

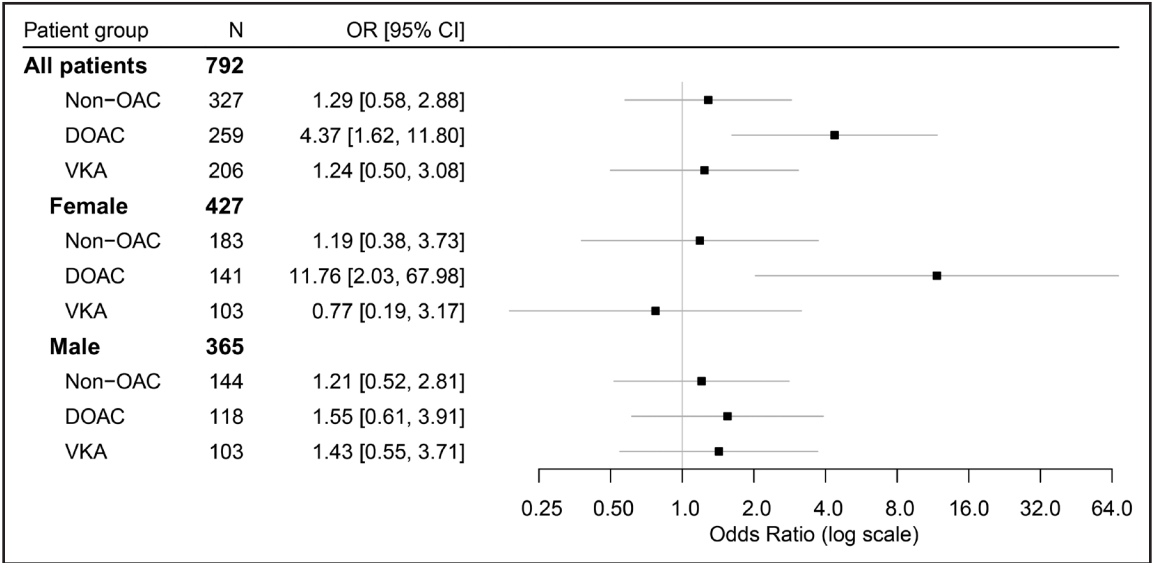


Figure 3. Odds ratios (OR) and 95% CIs for the association between ipsilateral stroke and vulnerable carotid plaques, estimated using a generalized linear mixed model and group-specific contrasts with emmeans.

To obtain the odds ratios for the direct oral anticoagulants (DOAC) and vitamin K antagonists (VKA) groups, the main effect of stroke ipsilateral (in the reference group no oral anticoagulation [non-OAC]) and the respective interaction effect (stroke ipsilateral×DOAC or ×VKA) were multiplied. Analyses were performed in the overall cohort and stratified by sex.

as potential sources of embolism.²⁴ In a study of 1406 patients with unilateral Embolic Stroke of Undetermined Source in the anterior circulation, vulnerable plaques were twice as common in the ipsilateral carotid artery compared with the contralateral side.⁴² To date, the prevalence of nonstenosing, vulnerable plaques in breakthrough strokes has not been systematically assessed. In our study, 34% of patients with unilateral ischemia in the carotid-territory and without carotid stenosis had ipsilateral vulnerable plaques compared with only 5% on the contralateral side. Patients with breakthrough strokes taking a DOAC had a 4-fold increase in the odds of ipsilateral carotid-territory ischemia. Previous CTA studies in patients with stroke demonstrated a predominance of vulnerable plaque features such as the rim sign on the side ipsilateral to brain infarction.^{18,43,44} The association between vulnerable plaques and ipsilateral brain ischemia was particularly pronounced in women in our study. Sex differences in carotid plaque stability, particularly in postmenopausal women, may explain the strong association between ipsilateral vulnerable plaques and carotid-territory ischemia observed in women anticoagulated with DOACs.^{45–47}

The best preventive management in patients after breakthrough strokes is currently unknown. Studies suggest that switching from one OAC to another may not be effective, and that addition of antiplatelet medication to OAC may even result in harm.^{2,4,48,49} Ongoing studies examine whether the addition of mechanical devices to OAC may improve the prevention of recurrent stroke after breakthrough stroke.² Two trials compare interventional left atrial appendage occlusion plus OAC against OAC alone (ELAPSE [Early Left Atrial Appendage Closure After Stroke in Atrial Fibrillation]; <https://www.clinicaltrials.gov>; Unique identifier: NCT05976685; LAAOS-4 [Left Atrial Appendage Occlusion Study IV]; <https://www.clinicaltrials.gov>; Unique identifier: NCT05963698). Bilateral transcatheter implantation of carotid filters to capture large emboli is an alternative approach which is in the early phase of investigation.⁵⁰ Although these approaches may reduce the risk of cardioembolism after breakthrough strokes, our data suggest that attention should also be paid to the prevention of embolism from carotid atherosclerotic plaques.

Strengths and Limitations

Our study has several strengths. It used data from a large, prospectively collected multicenter cohort of strokes related to AF that concurrently enrolled patients with and without anticoagulation. The use of a neuroradiologic core lab analysis ensured reliable and unbiased assessment of plaque morphology and stenosis using prespecified criteria. Several limitations warrant consideration. Participation in the RASUNOA-Prime registry demanded informed consent which may have introduced a selection bias. CTA images were performed in the acute setting as indicated by the local treating physician. While this may

be another source of selection bias, the overall study and the present CTA cohort were largely comparable. Patients with CTA had a shorter median symptom-onset to admission time and had higher stroke severity according to the National Institutes of Health Stroke Scale score (Table S2). Anticoagulation adherence and anticoagulation quality was not addressed in the RASUNOA-Prime study. Statin use, renal dose adjustment, INR control in VKA-treated patients, and the intensity of some vascular risk factors (eg, LDL [low-density lipoprotein] cholesterol) were not captured. CTA offers high spatial resolution for features like rim sign, ulceration, and soft-plaque thickness, but it is less sensitive than vessel wall magnetic resonance imaging for detecting intraplaque hemorrhage and lipid-rich necrotic core. Nevertheless, the rim sign and soft-plaque thickness are considered robust indicators of vulnerability.⁴¹ The contribution of insufficient anticoagulation by insufficient dosing or reduced adherence could not be systematically assessed in our study. In addition, we analyzed the vascular sides of the patients separately and modeled the presence of vulnerable plaques as the outcome. However, with this approach, the association with the ipsilateral ischemia side in our models is predominantly driven by the subgroup of patients with unilateral plaques. Finally, all subgroup and interaction analyses, including the sex-specific findings should be considered exploratory and hypothesis-generating only and were not adjusted for multiple comparisons.

Conclusions

Our results suggest that both stenosing carotid atherosclerosis and nonstenosing vulnerable carotid plaques are associated with breakthrough strokes in patients with AF. Longitudinal studies are needed to better understand the impact of atherosclerosis on stroke recurrence in patients with breakthrough stroke.

ARTICLE INFORMATION

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Author Contributions

R. Veltkamp conceived and initiated the study. D. Kinzler, A. Heeger, and A.W. Veltkamp performed central radiological adjudication and imaging analyses. Dr Hellwig and A. Hüsing conducted statistical analyses. A.W. Veltkamp and R. Veltkamp drafted the first version of the manuscript. All authors critically revised the manuscript for intellectual content and approved the final version.

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Disclosures

This study followed the EQUATOR Reporting Guideline (Enhancing the Quality and Transparency of Health Research) and STROBE checklist (Strengthening the Reporting of Observational Studies in Epidemiology) for cohort studies. ChatGPT (GPT-4.1 and GPT-5; OpenAI, San Francisco, CA; used September to December 2025) was used to assist for language editing of the manuscript text. The authors reviewed and edited all artificial intelligence–assisted content and take full responsibility for the accuracy and integrity of the final manuscript. Dr Purucker has received consultation fees and travel expenses from Abbott, Akcea, Bayer, Boehringer Ingelheim, Daiichi Sankyo, and BMS/Pfizer, and he reports grants from the Federal Joint Committee within the Innovation Fund. P. U. Heuschmann reports research grants from, University Hospital Heidelberg (within RASUNOA-Prime study [Registry of Acute Stroke Under Novel Oral Anticoagulants–Prime]; supported by an unrestricted research grant to the University Hospital Heidelberg from Bayer, BMS, Boehringer Ingelheim, Daiichi Sankyo); as well as from the German Ministry of Research and Education, Federal Joint Committee (G-BA), European Union, German Heart Foundation, German Research Foundation, Bavarian State, German Cancer Aid, Robert-Koch-Institute, outside the submitted work. R. Veltkamp reports research support from European Union's Horizon 2020 research and innovation program under grant agreement No 754517, and from Bayer, BMS-Pfizer, Boehringer Ingelheim, Daiichi Sankyo, Medtronic, Biogen. Honoraria for consultancies and lectures for Astra Zeneca, Bayer, BMS-Pfizer, Javelin, Portola, and he is an investigator of the Imperial BRC. The other authors report no conflicts.

Supplemental Material

Tables S1–S8
Figure S1
STROBE Checklist

APPENDIX

Collaborators and members of the RASUNOA-Prime study (Registry of Acute Stroke Under Novel Oral Anticoagulants–Prime) group listed in the following Appendix are indexed in PubMed. Local investigators and collaborators from centers who enrolled at least 1 patient in the RASUNOA-prime ischemic stroke study are listed.

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